SQL ASSIGNMENT 14

Subject: Database Management Systems (SQL) **Topic:** Python & SQLite Integration, Pandas DataFrames & ETL Automation **Status:** Completed

TASK 1

Install the `sqlite3` module in Python and write a script to create a new database called `foodie.db` with a table `Restaurants` (id, name, cuisine, rating).

```
import sqlite3 conn = sqlite3.connect('foodie.db') cursor = conn.cursor() cursor.execute('''
CREATE TABLE IF NOT EXISTS Restaurants ( id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT NOT NULL,
cuisine TEXT NOT NULL, rating REAL NOT NULL ) ''') conn.commit() conn.close()
```

TASK 2

Insert three sample restaurants into `Restaurants` and fetch all restaurants with rating > 4.0, printing their names.

```
import sqlite3 conn = sqlite3.connect('foodie.db') cursor = conn.cursor() sample_data = [
('Truffles', 'Italian', 4.6), ('Punjab Grill', 'Indian', 4.2), ('Quick Bites', 'Fast Food', 3.8) ]
cursor.executemany('INSERT INTO Restaurants (name, cuisine, rating) VALUES (?, ?, ?)', sample_data)
conn.commit() cursor.execute('SELECT name FROM Restaurants WHERE rating > 4.0') top_restaurants =
cursor.fetchall() for row in top_restaurants: print(row[0]) conn.close()
```

TASK 3

Load all rows from `Restaurants` table into a Pandas DataFrame and display the top 2 rows using `head()`.

```
import sqlite3 import pandas as pd conn = sqlite3.connect('foodie.db') df =
pd.read_sql_query('SELECT * FROM Restaurants', conn) conn.close() print(df.head(2))
```

TASK 4

Add a column `delivery_charge` set to 50, and calculate `final_rating` as `rating + 0.1` if cuisine is 'Italian'.

```
df['delivery_charge'] = 50 # Conditional logic using lambda and apply df['final_rating'] =
df.apply( lambda row: row['rating'] + 0.1 if row['cuisine'] == 'Italian' else row['rating'], axis=1 )
print(df)
```

TASK 5

Automate a daily summary script that fetches restaurants with rating > 4.5 into a DataFrame and saves them to `top_rated_restaurants.csv`.

```
import sqlite3 import pandas as pd def export_top_restaurants(): conn =
sqlite3.connect('foodie.db') query = 'SELECT * FROM Restaurants WHERE rating > 4.5' df_top =
pd.read_sql_query(query, conn) conn.close() df_top.to_csv('top_rated_restaurants.csv', index=False)
print("Daily summary exported successfully to top_rated_restaurants.csv") export_top_restaurants()
```